Intention-to-Treat Evaluation of P2Y12 Inhibitors Following Stent Implantation for Acute Coronary Syndrome on Time to Recurrence or Death in Austria (2020–2022)

Daniela Dunkler[1], Michael Kammer[1], S Necdet Cevirme[1], Moritz Pamminger[1] and Susanne Strohmaier[2]

[1]Institute of Clinical Biometrics, Center for Medical Data Science, Medical University of Vienna, Vienna, Austria

[2]Department of Epidemiology, Center for Public Health, Medical University of Vienna, Vienna, Austria

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[Standard Protocol]

Introduction to the Coding Experiment

The study protocol below serves as the foundation for the coding experiment. You are asked to conduct the study and report the results based on the attached study protocol and workflow description using the accompanying datasets. Please be aware that the experiment consists of two study arms, i.e. participants receive two different versions of the study protocol which differ regarding the level of detail provided in various sections. If you are in the 'standard' arm you received a study protocol and workflow description as we often encounter in practice (e.g. ethics committee submissions), for the 'detailed' study protocol several decisions are pre-specified in more detail.

Importantly, the primary goal of this experiment is not to produce the 'correct' results. Rather, the aim is to capture the variation that arises when different individuals translate imprecisely worded study protocols into precise, executable code.

To ensure that the submitted code is easily readable and executable for the study team, we provide a set of coding guidelines and ask you to follow them as closely as possible.

In subsequent evaluations of the coding experiment we are planning to focus on different phases of the coding process, **data preprocessing, including reporting of descriptive analysis results** and the **statistical analysis addressing the primary study objective, including reporting of main results and prespecified diagnostics** as outlined in the study protocol.

However, if you have alternative suggestions for any part of the realization of the SAP, please briefly sketch out the alternative steps in a dedicated separate file and ideally implement the alternative pathway as well. After completing the coding tasks, you will be invited to fill out a short questionnaire about your experience implementing the statistical analysis plan. Throughout the experiment we also encourage you to keep a short log documenting experiences, including any steps you found especially difficult or particularly easy, as well as the time you spent on each part of the analysis. We will invite participants who evidently attempted to complete all tasks to be co-authors of a publication of the results.

Please note that this experiment is not conducted anonymously. This is to facilitate communication and enable follow-up questions about the submitted code if necessary. Please also be aware that if several colleagues at your institution participate in the coding experiment not everybody necessarily was allocated to the same study arm. We kindly ask you to not share the different study protocol versions within your institutions.

Thanks again for your participation!

Guidelines for Coding, File Formats and Submission

We encourage you to write code that is easy for us to read and to execute. The following guidelines are intended to support this effort.

Programming language

Please write your code in R.

Data availability

A link and respective password to download all necessary files and information to conduct the coding experiment were made available via email after successful registration. If not, please contact us at `topstats2023@gmail.com`.

Submission Platform

The code and results can be uploaded via the ownCloud instance of the Medical University of Vienna LINK. The necessary password was made available via email after successful registration. If not, please contact us at `topstats2023@gmail.com`.
Multiple submissions are possible, but please only submit code when you are finished with your work, and only provide updates when it is absolutely necessary.

Folder Structure

Please submit a zip-compressed folder with a name that comprises your name and the submission date. Any potential updates should follow the same naming structure. The contents of the folder should be as follows:

```
README.md
LOG.md
ALTERNATIVE.md
data/
    └── cleaned_data.csv
code/
    ├── preprocessing.R
    └── analysis.R
output/
    ├── results.Rmd
    └── figures/
```

Note: See *File Formats* for possible alternative file-types.

Required Folders

- `data`: Contains the preprocessed data resulting from your code which should be ready to conduct the the main analysis
- `code`: Contains all your code files.
- `output`: Contains all analysis results generated by your code. It also contains a subfolder `figures` that comprises all figures generated by your code.

Required Files

Please include the following files in your folder and use them as outlined below. Please feel free to add additional files, for example for helper functions, as you see necessary to keep your codebase clean and understandable.
Avoid to upload the provided source data described in section 3, but upload only the analysis-ready data.

- `README.md`: A brief file describing the content of the submission, R version used, and package dependencies. For example, you can use the R function 'sessionInfo()' to do that. It should also outline how your scripts are to be called or used to replicate your results and data files. Please document here anything that another person would need to know to replicate the results using your code.
- `LOG.md`: A file reflecting on your coding experiences, including any steps you found especially difficult or particularly easy, as well as the time you spent on each part of the analysis.
- `preprocessing.R` (or .Rmd): The script used to clean and prepare the data.
- `cleaned_data.csv` (or .rds): The preprocessed dataset
- `resultsdescriptive.Rmd`: A summary of the descriptive analysis as described in the statistical analysis plan. Ideally, it would be an R Markdown file.
- `analysis.R` (or .Rmd): The script performing the main analysis.
- `resultsmain.Rmd`: A summary of the main findings as described in the statistical analysis plan. Ideally, it would be an R Markdown file.

Optional File(s) Describing Alternative SAP Realizations

If you would have preferred alternative decisions at one or several steps of the implementation of the SAP, please add the following file(s).

- `ALTERNATIVE.md`: Briefly describe your preferred alternative implementation of the SAP. Please indicate clearly at which step(s) in the SAP you would have diverted. In the optimal case, this file is accompanied by the actual code implementation of your alternative suggestion in separate R or Rmd file(s).

File Formats

- **Code**: '.R' or '.Rmd' (if using R Markdown)
- **Data**: '.csv' (please use the defaults of R's 'write.csv()' function to export data)
- **Results**: '.Rmd', or '.pdf' for plots
- **Documentation and Log**: '.Rmd', '.md' or '.txt'

Documentation

Please comment in your code when you...

- used the support of AI (please start the comment with "Comment-AI"),
- used the support of a colleague ("Comment-colleague"),
- had to clarify specific parts with the study team ("Comment-studyteam"), or

- had to make a conscious decision during coding, which was not elaborated in the statistical analysis plan ("Comment-decision").

Additional Recommendations

- Use relative paths in your code (‘`./data/`‘, ‘`./output/`‘) to facilitate reproducibility.

FAQ

A 'Frequently Answered Questions' (FAQ) document is made available via the ownCloud instance of the Medical University of Vienna and continually updated. We encourage participants to read the FAQ. If one of the participants has a question of public interest, the question and the answer will be posted in the FAQ. If an update is considered relevant to all participants, you will be notified.

Contact us

If content-related ambiguities hinder the coding process, you can contact us at `topstats2023@gmail.com`.

Submission Date

Please upload all your files latest by March 1st 2026.

Study Protocol including the Statistical Analysis Plan

Intention-to-Treat Evaluation of P2Y12 Inhibitors Following Stent Implantation for Acute Coronary Syndrome on Time to Recurrence or Death in Austria (2020–2022)

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1 Introduction

Cardiovascular disease remains the leading cause of death worldwide, despite significant advances in diagnosing and treating acute coronary syndromes (ACS)—a group of conditions caused by reduced blood flow to the heart, such as heart attacks. One important element in the treatment of ACS is antithrombotic therapy, which helps prevent blood clots. This therapy plays a central role during and after procedures like percutaneous coronary intervention, especially when a stent (a small mesh tube) is implanted to keep blocked arteries open.[1]

In Austria, three **P2Y12 inhibitors**[2] — **clopidogrel, prasugrel, and ticagrelor** — are available as oral medications. These drugs help reduce the risk of further heart events by preventing platelets in the blood from clumping together. According to European treatment guidelines (Roffi et al. 2016), one of these medications should be prescribed as soon as possible after an ACS event and continued for 12 months—unless a patient has specific risks, such as a high chance of bleeding. However, in actual clinical practice, decisions about which drug to use, when to start treatment, and how long to continue it may vary depending on local expertise and the results of the heart procedure.

In this study, we will analyze healthcare reimbursement data from the Austrian Health Insurance Fund (Österreichische Gesundheitskasse, ÖGK) [3]. Our goal is to better understand how P2Y12 inhibitors are used in a real-world setting for individuals who were hospitalized for ACS and received a stent. Specifically, our objective is to examine how different treatment choices relate to the risk of a first recurrence of ACS or death.

[1] *Disclaimer:* We present the medical background here in simplified form and focus on those aspects that are relevant for understanding the study.

[2] P2Y12 refers to a type of purinergic receptor. These receptors play a crucial role in platelet activation and aggregation, which are essential processes in blood clotting. Drugs that inhibit P2Y12, such as clopidogrel, prasugrel, and ticagrelor, are commonly used to prevent thrombus formation in patients with cardiovascular diseases, particularly those who have undergone procedures like stent placement.

[3] *Disclaimer:* For data protection reasons we synthesized the reimbursement data we had received to mimic the original information and data formats.

2 Research aim

The primary aim of this study is to quantify the **intention-to-treat (ITT) effect of the choice of therapy on time until first recurrence of ACS or death** in individuals who were hospitalized for stent implantation after ACS in Austria between October 1st 2020 to December 31st 2022.

2.1 Population of interest

The population of interest consists of all individuals who were hospitalized for ACS and received a stent in Austria between October 1st 2020 to December 31st 2022 and who redeemed at least one P2Y12 inhibitor—clopidogrel, prasugrel, or ticagrelor—after the stent implantation. Included individuals are treatment-naive (see exclusion criteria).

2.2 Index Hospitalization

Patients hospitalized for a stent may have several distinct hospital stays, which may be very close in time. These 'close' hospital stays will be treated as a single, 'pooled' hospital stay, i.e., *the index hospitalization*. It is assumed, that hospitalizations within 30 days of each other, stem from the same ACS event and hence, will not be treated as newly recurred ACS. The pooling of hospitalizations may need to be conducted iteratively.

2.3 Eligibility criteria

Inclusion criteria are

- age at least 18 years or older at beginning of index hospitalization
- hospitalization admitted on or after October 1st 2020 and discharged by December 31st 2022
- a hospitalization with the diagnosis of ACS defined by the following ICD-10 codes[4]

[4] **ICD-10 codes** are standardized alphanumeric codes maintained by the WHO and used worldwide to classify and record diseases, symptoms, and medical procedures. Each code corresponds to a specific diagnosis or health condition, allowing for consistent documentation across healthcare systems, billing, and epidemiological research. ICD-10 codes are organized hierarchically into four levels:

1. **Chapter level**: Broad categories based on body systems or disease types, e.g., *Chapter IX: Diseases of the circulatory system (I00–I99)*
2. **Block level**: Groups of related diseases within a chapter, e.g., *I20–I25=Ischaemic heart diseases*
3. **Category level**: 3-character codes for specific disease groups, e.g., *I21=Acute myocardial infarction* (heart attack)
4. **Subcategory level**: 4-character codes offering further detail (site, severity), e.g., *I211=Acute transmural myocardial infarction of inferior wall*

English version: https://icd.who.int/browse10/2019/en, German version: https://klassifikationen.bfarm.de/icd-10-gm/kode-suche/htmlgm2025/index.htm).

  - Unstable angina pectoris (uAP): I20.0, I20.1, I20.8, I20.9
  - Non ST Elevation Myocardial Infarction (N-STEMI): I21.4, I21.9, I22.0, I22.1, I22.2, I22.8, I22.9, I24.0, I24.8, I24.9
  - ST Elevation Myocardial Infarction (STEMI): I21.0, I21.1, I21.2, I21.3

  The accompanying file `"ICD10forACS.csv"` contains the list of relevant ICD-10 codes.

- a hospitalization with a stent implantation as indicated by one of following Medical Individual Service codes (Medizinische Einzelleistungen, MEL) codes[5]
  - Coronary stent implantation: DD050
  - Implantation of a drug-eluting stent in the coronary vessels: DD060
  - Implantation of a fully bioresorbable stent in the coronary vessels: XN120

- redemption of a P2Y12 inhibitor between the start of the index hospitalization and 30 days after discharge, indicated by the following ATC codes[6]:
  - clopidogrel: B01AC04
  - prasugrel: B01AC22
  - ticagrelor: B01AC24

Exclusion criteria are

- diagnosis of ACS in the 12 months before the index hospitalization.
- stent implantation in the 12 months before the index hospitalization.
- redemption of one of the three P2Y12 inhibitors in the 12 months before the start of the index hospitalization
- observation period shorter than 30 days after discharge of the index hospitalization
- death within 30 days after discharge of the index hospitalization

[5] **MEL** codes are part of the Federal Ministry of Labour, Social Affairs, Health, Care and Consumer Protection (BMSGPK) Service Catalogue, the binding basis for uniform national service documentation of medical interventions in Austria. The Service Catalogue includes both inpatient and outpatient procedures.

[6] **ATC** codes are a standardized classification of drugs based on the organ/system acted upon, therapeutic use, and chemical characteristics. See https://atcddd.fhi.no/atc_ddd_index/. The system is hierarchical and includes five levels. For example, the ATC for the P2Y12 inhibitor *clopidogrel* is *B01AC04* and reads like

- the chemical subgroup/substance **B01AC04** corresponds to *clopidogrel* – used to prevent blood clots in ACS
- the pharmacological subgroup **B01AC** corresponds to *Platelet aggregation inhibitors excl. heparin*
- the therapeutic subgroup **B01** corresponds to *Antithrombotic agents*
- the anatomical main group **B** corresponds to *Blood and blood forming organs*

2.4 Primary aim

- **Estimand:** Intention-to-treat effect comparing survival probabilities across different drug prescriptions after ACS with stent
- **Target Population:** individuals (at least 18 years) with percutane coronar intervention (i.e., stent) due to an ACS between 2020 and 2022 hospitalized in Austria who redeemed a relevant oral inhibitor of platelet aggregation within 30 days of hospital discharge
- **Exposure contrast:** clopidogrel, prasugrel, or ticagrelor
- **Outcome:** time to first re-admission for ACS or death (composite outcome), administrative censoring on December 31, 2024
- **Outcome measures:**
  - survival curves per drug
  - 3-years restricted mean survival time (RMST) per drug
- **Time 0:** 30 days after the end of the pooled index hospitalization
- **Follow-up:** up until 3 years after time 0
- **Type of analysis strategy:** Due to the lack of certain important confounders, causal interpretation will be hampered, however, we follow the causal road-map to inform our analysis strategy.

3 Description of the available data sources

We will utilize demographic data, prescription data, individual reimbursement data and information on hospital discharges with primary diagnoses. The database consists of five *.csv* files called "hospitalstay_all", "hospitalstay_stent", "mel_stent", "patients", and "redemptions". Included are all individuals who received a stent in Austria between October 1st 2020 to December 31st 2022.

The variable *id* serves as the unique identifier for every individual and is included in all five files. Some individuals may have had more than one hospitalization during the observation period. The variable *stent_id* links different hospital stays within an individual.

3.1 File *patients*

This file includes data on 125,000 individuals who have received at least one stent in the period from October 1st 2020 to December 31st 2024. The file *patients* contains information on the variables found in Table 1.

Table 1: Variables in the file *patients*.

Variable	**Description**	**Type**	**Comment**
id	pseudonymized patient identifier	number	linking attribute
sex	sex of individual	text	f(emale), m(ale)
birth_year	year of birth of individual	number	year
death_date	date of death of individual	date	dd/mm/yyyy

3.2 File *hospitalstay_stent*

This file contains data on 138,893 hospital stays with a stent implantation during the period October 1st 2020 to December 31st 2024, referring to the 125,000 patients from the file 3.1 *patients*. Each hospital stay includes up to three main diagnoses. Diagnoses are given as ICD-10 codes (International Statistical Classification of Diseases and Related Health Problems, 10th Revision). In the file, either category-level (3-character) or subcategory-level (4-character, **without the dot**) codes are recorded. Table 2 lists the variables found in *hospitalstay_stent* contains the variables found in .

Table 2: Variables in the file *hospitalstay_stent*.

Variable	**Description**	**Type**	**Comment**
id	pseudonymized patient identifier	number	linking attribute
stent_id	sequential number of hospital stay per patient	number	linking attribute to *mel_stent*
stent_begin_date	start of hospital stay	date	dd.mm.yyyy
stent_end_date	end of hospital stay	date	dd.mm.yyyy
diag1	diagnosis 1 for hospital stay	text	ICD10 code
diag2	diagnosis 2 for hospital stay	text	ICD10 code
diag3	diagnosis 3 for hospital stay	text	ICD10 code

3.3 File *mel_stent*

This file contains data on 138,896 implanted stents, which occured during hospital stays recorded in the file 3.2 *hospitalstay_stent*. A stent implantation occurs on a specific day during a hospital stay. During one stenting procedure at a specific day, an individual can receive up to six different stents.

The Austrian Health Insurance Fund bills medical services using so-called **Medical Individual Service codes (Medizinische Einzelleistungen, MEL)**. This coding system encompasses all billable medical services. Stents used for coronary implantation are coded using the MEL codes found in Table 3.

Table 3: Overview of MEL codes related to coronary interventions. The accompanying file `MEL.csv` contains the list of MEL codes mentioned in the eligibility criteria section 2.3.

MEL	Description
DD050	Implantation of a stent in the coronary vessels
DD060	Implantation of a drug-eluting stent in the coronary vessels
XN120	Implantation of a fully bioresorbable stent in the coronary vessels
DD010	Catheter angiography of the coronary vessels
DD040	PTCA – Percutaneous transluminal coronary angioplasty
DD070	Rotational angioplasty of the coronary vessels
DD075	Percutaneous transluminal lithotripsy of a coronary artery
DD080	Intracoronary thrombus aspiration
DD090	PTCA with cutting balloon
DD100	Percutaneous transluminal atherectomy of a coronary artery
DD105	Percutaneous recanalization of chronically occluded coronary arteries
DD110	Distal protection during coronary interventions

The file *mel_stent* contains the variables found in Table 4.

Table 4: Variables in the file *mel_stent*.

Variable	Description	Type	Comment
id	pseudonymized patient identifier	number	linking attribute
stent_id	sequential number of hospital stay per patient	number	linking attribute to *hospitalstay_stent*
stent_date	date of intervention (stent placement)	date	dd/mm/yyyy
mel1	MEL 1 on the implantation date	text	see Table 3
mel2	MEL 2 on the implantation date	text	see Table 3
mel3	MEL 3 on the implantation date	text	see Table 3
mel4	MEL 4 on the implantation date	text	see Table 3
mel5	MEL 5 on the implantation date	text	see Table 3
mel6	MEL 6 on the implantation date	text	see Table 3

3.4 File *hospitalstay_all*

This file contains data on 353,547 hospital stays. **All** hospital stays with a discharge date from September 1st 2019 to December 31st 2024 are included, also those from the file 3.2 *hospitalstay_stent*. These hospital stays can be identified via the variables *id*, *cohosp_begin_date*, and *cohosp_end_date*. Each individual from the file 3.1 *patients* has at least one hospital stay. Each hospital stay includes up to three main diagnoses. Diagnoses are given as ICD-10 codes. In the file, either category-level (3-character) or subcategory-level (4-character, **without the dot**) codes are recorded.

The file *hospitalstay_all* contains the variables found in Table 5.

Table 5: Variables in the file *hospitalstay_all*.

Variable	Description	Type	Comment
id	pseudonymized patient identifier	number	linking attribute
cohosp_begin_date	start of hospital stay	date	dd.mm.yyyy, linking attribute to hospitalstay_stent
cohosp_end_date	end of hospital stay	date	dd.mm.yyyy, linking attribute to hospitalstay_stent
diag1	diagnosis 1 for hospital stay	text	ICD10 code
diag2	diagnosis 2 for hospital stay	text	ICD10 code
diag3	diagnosis 3 for hospital stay	text	ICD10 code

3.5 File *redemptions*

This file contains 7,454,688 redeemed prescriptions. Most individuals (from the file 3.1 *patients*) have redeemed at least one medication prescription between December 1st 2016 and December 31st 2024. Medications are recorded using ATC (Anatomical Therapeutic Chemical Classification System) codes. The file *redemptions* contains the variables found in Table 6.

Table 6: Variables in the file *redemptions*.

Variable	Description	Type	Comment
id	pseudonymized patient identifier	number	linking attribute
redemption_date	prescription redemption date	date	dd.mm.yyyy
ATC	ATC code	text	5th level ATC code

4 Variables in the main analysis

4.1 Exposure contrast

As this is an ITT analysis, the exposure of interest is defined as the first redeemed P2Y12 inhibitor (either clopidogrel, prasugrel or ticagrelor).

4.2 Adjustment covariates

Adjustment covariates include

- Age in years
- Sex (male or female)
- Year of the start of the (pooled) index hospitalization
- Number of days spent in the hospital during the (pooled) index hospitalization
- Number of days spent in the hospital during the eligibility period [7]

[7] *i.e. 365 days before (pooled) index hospitalisation start, see Figure 1 for a graphical definition of the eligibility period*

- Number of relevant ICD-10 diagnoses during the (pooled) index hospitalization. For the exact codes used see Table 8.
- Number of ICD-10 diagnoses during the eligibility period
- Total number of redeemed medications during the eligibility period
- Binary indicators for different groups of redeemed medications during the eligibility period. For the exact groupings of medications see Table 7.

Table 7: Groups of comedication used as adjustment covariates.

ATC	**Description**
A10A	Insulins and their analogue
A10B	Blood glucose lowering drugs, excluding insulins
B01	Antithrombotic agents
R01	Nasal preparations
R03	Anti-asthmatics
C01	Cardiac Therapeutics
C02	Antihypertensives
C03	Diuretics
C07	Beta blocking agents
C08	Calcium channel blockers
C09	Agents acting on the renin-angiotensin system
C10	Lipid modifying agents

Table 8: Groups of comorbidities used in the descriptive analysis.

Disease	**ICD**
Arrhythmia	I440, I441, I442, I49, I490, I491, I492, I493, I494, I495, I498, I499, I470, I471, I479
Atrial Fibrillation	I48, I480, I481, I482, I483, I484, I489
Cancer	C00, C01, C02, C03, C04, C05, C06, C07, C08, C09, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75
Congestive Heart Failure	I50, I500, I5000, I5001, I5002, I5003, I5004, I5005, I501, I5011, I5012, I5013, I5014, I5019, I509, I42, I420, I421, I422, I423, I424, I425, I426, I427, I428, I4280, I4288, I429
Coronary Artery Disease	I220, I221, I228, I229, I240, I241, I252, I2520, I2521, I2522, I2529, I253, I254, I255, I256, I258, I259, I200, I201, I208, I209, I21, I210, I211, I212, I213, I214, I219, I248, I249, I250, I251, I2510, I2511, I2512, I2513, I2514, I2515, I2516, I2519
Diabetes	E100, E1001, E101, E1011, E102, E1020, E1021, E103, E1030, E1031, E104, E1040, E1041, E105, E1050, E1051, E106, E1060, E1061, E107, E1072, E1073, E1074, E1075, E108, E1080, E1081, E109, E1090, E1091, E110, E1101, E111, E1111, E112, E1120, E1121, E113, E1130, E1131, E114, E1140, E1141, E115, E1150, E1151, E116, E1160, E1161, E117, E1172, E1173, E1174, E1175, E118, E1180, E1181, E119, E1190, E1191, E120, E1201, E121, E1211, E122, E1220, E1221, E123, E1230, E1231, E124, E1240, E1241, E125, E1250, E1251, E126, E1260, E1261, E127, E1272, E1273, E1274, E1275, E128, E1280, E1281, E129, E1290, E1291, E130, E1301, E131, E1311, E132, E1320, E1321, E133, E1330, E1331, E134, E1340, E1341, E135, E1350, E1351, E136, E1360, E1361, E137, E1372, E1373, E1374, E1375, E138, E1380, E1381, E139, E1390, E1391, E140, E1401, E141, E1411, E142, E143, E1430, E144, E1441, E1450, E1451, E1461, E147, E1472, E1475, E148, E1481, E1490, E1491, R730, R739
Hemorrhagic stroke	I61, I610, I611, I612, I613, I614, I615, I616, I618, I619
Hyperlipidemia/Secondary Prophylaxis	E78, E781, E782, E783, E784, E785, E786, E788, E789
Hypertension	I1000, I1001, I101, I1010, I1011, I109, I1090, I1091, I110, I1101, I119, I1190, I1191
Immunosuppressive Disease	B20, B21, B22, B23, B24
Immunosuppressive Status	D90
Immunosuppressive Therapy	Z940, Z941, Z942, Z943, Z944, Z946, Z9481, Z9488, Z949
Ischemic stroke	G45, G450, G451, G452, G453, G454, G458, G459, G468, I63, I630, I631, I632, I633, I634, I635, I636, I638, I639
Kidney Disease	N17, N170, N171, N172, N178, N179, N181, N182, N183, N184, N185, N188, N1889, N189, N19
Liver Disease	K70, K71, K72, K73, K74, K75, K76, K77
Obesity	E6606, E6607, E6608, E6616, E6617, E6618, 6626, E6627, E6628, E6686, E6687, E6688, E6696, E6697, E6698
Peripheral Artery Disease	I7020, I7021, I7022, I7023, I7024, I7025, I7026, I7029, I708, I709
Pulmonary Disease	J440, J4400, J4401, J4402, J4403, J4409, J441, J4410, J4411, J4412, J4413, J4419, J448, J4480, J4481, J4482, J4483, J4489, J449, J4490, J4491, J4492, J4493, J4499, J450, J451, J458, J459, I270, I2720, I2728, E84
Pulmonary embolism	I26, I260, I269

5 Statistical Analysis

Figure 1 summarizes the timeline and the periods relevant for the inclusion and exclusion criteria, as well as for the collection of covariates.

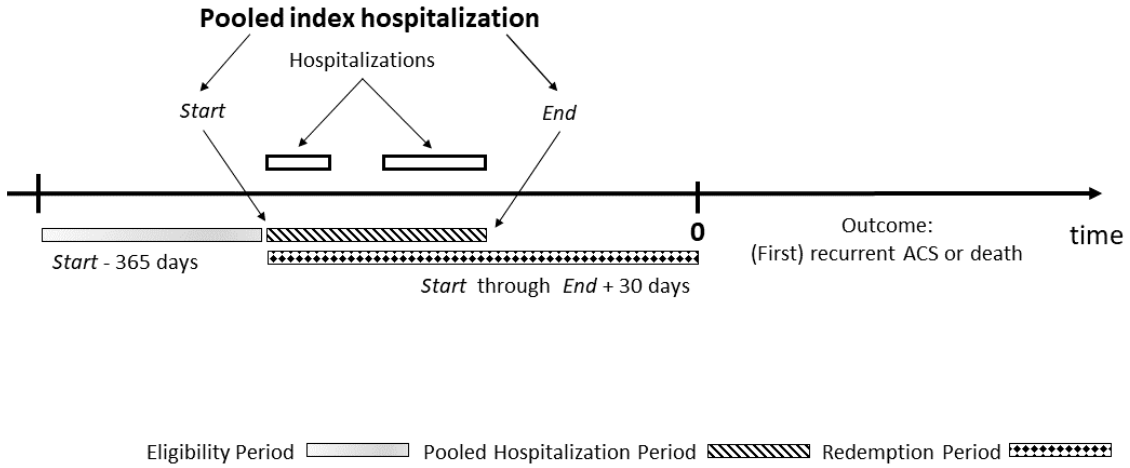


Figure 1: Time frame of the study: The study includes individuals hospitalized for acute coronary syndrome (ACS) who underwent stent implantation. Multiple stenting procedures may occur during a single hospitalization. Recurrent hospitalizations within short time spans between subsequent visits (30 days) are aggregated into a single, pooled hospitalization episode.
In the eligibility period (i.e. 365 days before the start of the index hospitalization), eligible individuals must not have had any ACS-related hospitalizations or redeemed prescriptions for P2Y12 inhibitors.
Individuals are included if they redeem at least one prescription for a P2Y12 inhibitor within the redemption period.
Time zero is defined as 30 days after the end of the pooled index hospitalization. The primary outcome is a composite endpoint of the first recurrent ACS hospitalization or death.

5.1 Preprocessing

Data cleaning. As these are reimbursement data, the data quality is assumed to be very good.

Definition of pooled index hospitalization. Consecutive hospitalizations separated by 30 days are combined into a single pooled index hospitalization, defined from the start of the first admission to the end of the last admission within this window. The pooling of hospitalizations may need to be conducted iteratively.

Handling of missing or faulty data. Individuals with missing or faulty data on sex and age are excluded.

5.2 Descriptive analysis

A table will present the distribution of the three treatment groups at the first prescription following index hospitalization. In a second table baseline characteristics, i.e., all adjustment covariates, will be summarized for the entire cohort and stratified for each treatment group. Continuous variables will be summarized as means with standard deviations or medians with interquartile ranges, depending on distribution, while categorical variables will be reported as counts and percentages. An unstratified Kaplan-Meier plot with 95% confidence intervals will illustrate time to first re-admission for ACS or death. The reverse Kaplan–Meier estimator (status indicator is reversed) is used to estimate the distribution of observation times with the median follow-up time (overall and separately for the three medication groups) reported.

5.3 Analysis for primary aim

A Kaplan-Meier plot stratified by the three treatment groups will be created. To answer the primary research question marginal survival curves will be compared between patients redeeming either clopidogrel, prasugrel, and ticagrelor at the first occasion after the index hospitalization. To estimate these marginal survival curves double robust standardization will be applied, which combines inverse probability of treatment weighting (IPTW) and the g-formula following these steps:

1. fit a multinomial model for treatment with clopidogrel, prasugrel or tigracelor using the covariates from 4.2 to obtain IPTWs. Further, use the marginal probability of the observed treatment to obtain stabilized IPTWs. Depending on their distribution the weights may be winsorized at the 95th percentile.

2. fit a Cox model for the primary outcome applying the weights from step 1 and additionally including the adjustment covariates in the model (double robustness)

3. apply regression standardization (g-formula) based on the model from step 2

5.4 Reporting

Assumptions and diagnostics

Basic model checking will be conducted and reported for completeness [8]. Diagnostics of un-stabilized and stabilized weights (e.g., distribution via boxplot and histogram / density, and absolute / relative frequencies of extreme weights > 10) will be reported to evaluate positivity and model adequacy. The proportional hazards assumption for the outcome model will be evaluated by standard model diagnostics through

[8] *Although such aspects would typically be discussed in greater detail, no further actions or revisions in response to potential misspecifications are required beyond this point, as this is beyond the scope of the experiment. As discussed in the beginning of this document, we welcome additional analysis in separate analysis and results file.*

Schoenfeld residuals.

Analysis results

The marginal survival curves for each treatment group will be plotted. From the marginal survival curves an estimate of the restricted mean survival time (RMST) at 3 years will be computed for each group.

6 Software

For data processing, analysis and visualizations the statistical programming language R is used.

7 Data protection

The coders will receive a synthesized version of the original data for analysis.

8 Abbreviations

- ACS – acute coronary syndrome
- ITT – intention-to-treat
- IPTW - inverse probability of treatment weighting
- MEL – Medical Individual Service codes (Medizinische Einzelleistungen)
- N-STEMI – non ST Elevation Myocardial Infarction
- ÖGK – Austrian Health Insurance Fund (Österreichische Gesundheitskasse)
- PTCA – percutaneous transluminal coronary angioplasty
- STEMI – ST Elevation Myocardial Infarction
- uAP – unstable angina pectoris

9 References

Roffi, M. et al. (2016). "2015 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: Task Force for the Management of Acute Coronary Syndromes in Patients Presenting without Persistent ST-Segment Elevation of the European Society of Cardiology (ESC)". In: *Eur Heart J* 37.3, pp. 267–315. DOI: 10.1093/eurheartj/ehv320.